

Acric Ferralsols:

Acric Ferralsols are deeply weathered tropical soils with a reddish to yellowish color caused by high iron and aluminum oxide accumulation.

They are named “Acric” because of their very low base saturation (<35%), which makes them strongly acidic and relatively infertile compared to Eutric Ferralsols. The soil has a deep, well-structured, clay-rich subsurface that is physically stable and resistant to erosion but chemically poor. These soils typically occur on well-drained uplands in humid tropical regions.

Drainage:

- **Wet season:** Well-drained; water infiltrates readily due to stable structure, but low permeability in deeper horizons may cause temporary ponding in micro-depressions.
- **Dry season:** Soil remains firm and friable; good internal drainage supports crop growth.
- **Landscape:** Undulating to hilly tropical uplands; rarely in low-lying areas.

Depth:

- Very deep (>150 cm); rooting is generally unrestricted physically, but nutrient limitations may reduce crop growth.

Workability:

- **When wet:** Plastic and sticky but manageable; does not smear like Vertisols.
- **When dry:** Firm and friable; easy to till and prepare seedbeds.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (10–20 cmol(+)/kg).
- **Base saturation:** Very low (<35%), making soils acidic and nutrient-poor.
- Phosphorus is highly fixed; nitrogen and organic matter are low. Potassium is also low.
- Iron and aluminum oxides dominate; micronutrients (Zn, Mn, Cu) may be insufficient.

pH:

- Strongly to moderately acidic (pH 4.0–5.5).

Management:

- Incorporate organic matter (manure, compost, crop residues) to improve fertility, structure, and water-holding capacity.
- Apply phosphorus fertilizers carefully, using slow-release rock phosphate with occasional superphosphate.
- Liming may be required to correct acidity and improve base saturation.
- Maintain vegetation cover or cover crops to reduce erosion.
- Avoid heavy machinery during wet periods to prevent compaction.
- Suitable crops include plantation crops such as cocoa, coffee, oil palm, rubber, and smallholder food crops with proper fertilization.